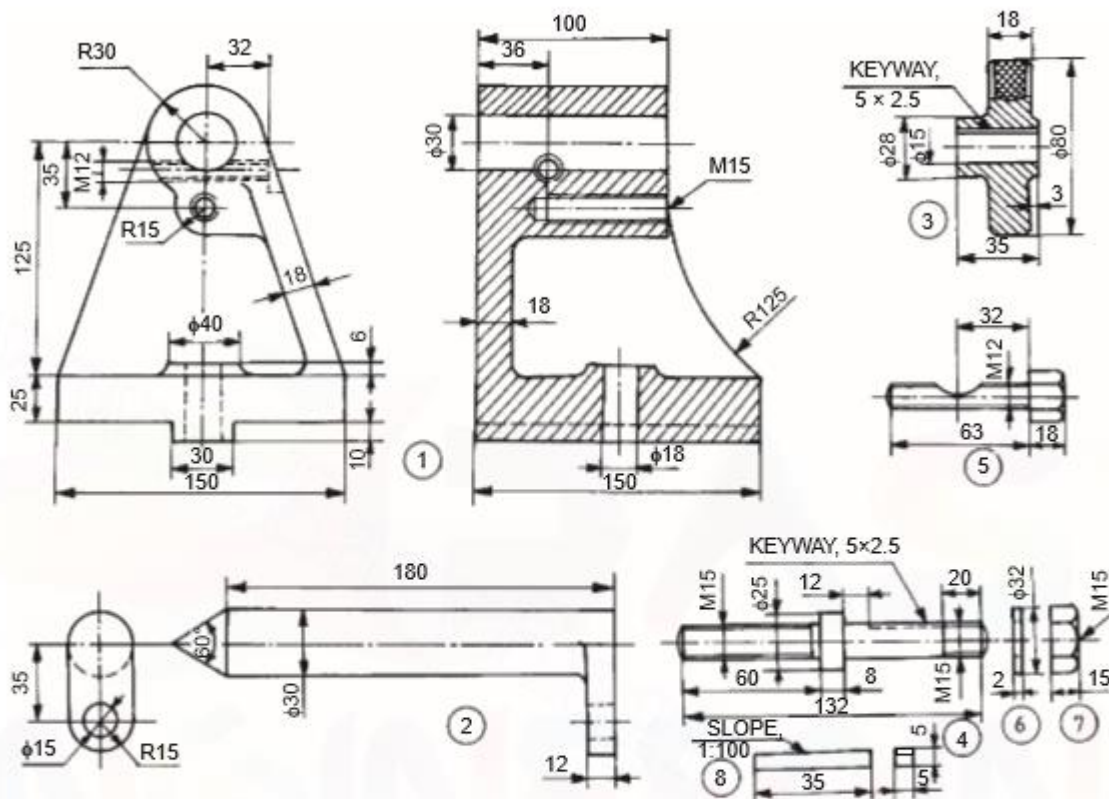


Milling machine tail-stock

Jobs requiring milling operations in relation to their axes of rotation, are usually supported between the centres of the dividing head and adjustable centre provided in the tail-stock. The parts of a milling machine tail-stock are shown in Fig. 18.19. This is similar to the lathe tailstock. The screw 4 is introduced into the threaded hole of body 1. Centre 2 is inserted into the body such that, the hole provided in it enters onto the screw 4. Hand wheel 3 is mounted on the screw 4, by using the key 8 and fixed in position by using washer 6 and nut 7. By operating the hand wheel, the centre can be given the required movement/fine adjustment, while clamping the job between the centres. After adjustment, the centre can be locked by the screw 5, which is introduced into the body, prior to the location of the centre in the body. Figure below shows the assembly drawing.

Assemble the parts of the milling machine tail-stock, shown in Fig. 18.19 and draw, (i) sectional view from the front and (ii) view from the right.



Parts list

Sl. No.	Name	Matl.	Qty.
1	Body	CI	1
2	Centre	Case hardened alloy steel	1
3	Hand wheel	Cast steel	1
4	Screw	MS	1
5	Screw	MS	1
6	Washer	MS	1
7	Nut	MS	1
8	Key	MS	

Revolving center

When long bars are machined on a lathe, they are supported on two centers. one of which is called a live centre and the other, a dead centre, fixed in the tail-stock. The live Centre fits into the main spindle and revolves with the work it supports. Because of the relative motion between the work piece and the dead centre in the tail-stock barrel, over-heating and wear of the centre takes place in the long run. To eliminate this, the dead centre is replaced with a live or anti-friction bearing centre, which revolves with the work like a live centre. Figure below shows the details of a revolving centre using antifriction bearings. The radial bearing 6 and thrust bearing 7 used in the design are meant for resisting the possible radial and axial loads respectively. The sleeve 4 is press fitted in the barrel 1 to provide end support to the centre 3. The sleeve is positioned in the barrel by the cover 5. Another cover 2 is fixed on the front side of the barrel by means of the screws 8 to retain the radial bearing in position. Assemble the parts of the revolving centre, shown in Fig. below and draw a half sectional view from the front.

